

QUALITY ASSURANCE (QA) TEST REPORT

Project Name

Voice Assistant Bot with Speech Recognition and Automation

1. Objective

The objective of this testing activity is to verify that the Voice Assistant Bot functions according to the requirements specified in the Project Requirement Document (PRD). Testing will ensure that all voice commands, automation features, task management operations, note management functions, database operations, and system responses work correctly and reliably.

2. Scope of Testing

The following modules are included in testing:

1. Voice Input & Speech Recognition
2. Command Processing
3. Task Management
4. Notes Management
5. System Automation
6. Web Automation

- 7. Information Services
- 8. Voice Response (Text-to-Speech)
- 9. Performance & Stability Testing

3. Test Environment

Parameter	Details
Operating System	Windows 11
Programming Language	Python
Microphone	Working Microphone Required
Internet Connection	Required for Speech Recognition
Browser	Chrome / Edge

4. Functional Test Cases

Test Scenario	Test Steps	Expected Result	Status
Assistant Startup	Run the program	Assistant should say "Assistant Started" and "Voice Assistant Active"	Pass
Create CSV Files	Run program for first time	task.csv and notes.csv should be created automatically	Pass

Voice Recognition (Normal Speech)	Say "Hello Assistant" clearly	Assistant should respond with greeting	Pass
Voice Recognition (No Speech)	Stay silent for 5 seconds	Program should display "No speech detected."	Pass
Voice Recognition (Unclear Speech)	Speak random noise or unclear words	Program should display "Could not understand audio."	Pass
Add Task	Say "Create task Complete project"	Task should be saved in task.csv with Pending status	Pass
Show Tasks	Say "Show tasks"	All saved tasks should be displayed	Pass
Complete Task	Say "Complete task Complete project"	Task status should change to Completed	Pass
Delete Task	Say "Delete task Complete project" and then "Yes"	Task should be removed from CSV	Pass
Delete Task Cancellation	Say "Delete task Complete project" and then "No"	Task should remain in CSV	Pass
Take Note	Say "Take note Meeting at 5 PM"	Note should be stored in notes.csv	Pass
Show Notes	Say "Show notes"	All notes should be displayed with ID and timestamp	Pass
Open Google	Say "Open Google"	Google should open in browser	Pass

Open YouTube	Say "Open YouTube"	YouTube should open in browser	Pass
Google Search	Say "Search Python tutorial"	Google search results should open	Pass
Time Query	Say "What is the time"	Current time should be spoken	Pass
Date Query	Say "What is today's date"	Current date should be spoken	Pass
Day Query	Say "What day is today"	Current day should be spoken	Pass
Help Command	Say "Help"	Supported commands list should be displayed	Pass
Stop Assistant	Say "Stop assistant"	Assistant should say goodbye and exit	Pass

5. Issues Found

Issue ID	Description	Status
1	complete_task() always says "Task marked as completed" even if the task does not exist	Fail
2	delete_task() always says "Task is deleted" even if the task does not exist	Fail
3	show_tasks() may crash if task.csv becomes empty or corrupted	Fail

4	show_notes() may raise an error if the CSV header is missing	Fail
5	User can say only "Create task" without a task name and an empty task gets stored	Fail
6	User can say only "Complete task" without a task name and the assistant still processes the request	Fail
7	User can say only "Delete task" without a task name and the assistant asks for confirmation	Fail
8	If internet is unavailable, Google Speech Recognition raises a RequestError	Needs handling (display proper error message)
9	If the microphone is unavailable or disconnected, microphone initialization may fail	Needs Handling (display proper error message)